Uniform Gap-Aware Quotient Laws

Price Envelopes, Stopped Safety, and a Ratio-Collapse Boundary

Liang Wang

July 2026

Abstract

The weak-mode quotient route has two distinct layers. RH-96 proves a local gap-weighted loss bound, while RH-102 proves that exact propagated debits can be stopped against endpoint slack. This paper supplies the missing uniform bridge between them.

For a positive compressed block

$$H = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix},$$

write $g = \alpha - \beta > 0$ for a retained-to-omitted spectral gap and $c = \|C\|_{\mathrm{F}}$. The local quotient price is

$$\ell = \frac{c^2}{g}.$$

If a propagated debit is bounded by $K\ell$, then N candidate quotients fit a stopped allowance ηS whenever

$$NK \sup_i \frac{c_i^2}{g_i} \leq \eta S.$$

We prove that all candidates are then accepted and the endpoint gate is preserved. If this uniform fit fails, the same theorem gives a safe stopped prefix: the first unaffordable candidate is rejected and the remaining chain uses full width. Thus uniform safety does not require a perpetual supply of quotients.

A power corollary makes the all-level target explicit. If $c_i = O(\sigma^\chi \operatorname{polylog})$, $g_i \gtrsim \sigma^\gamma / \operatorname{polylog}$, $K = O(\sigma^{-\kappa} \operatorname{polylog})$, and the endpoint slack is $\gtrsim \sigma^s / \operatorname{polylog}$, then the total price has signed decay exponent $2\chi - \gamma - \kappa$. The no-stop regime is obtained when that exponent reaches the slack exponent s, with the constant budget also fitting. A collapsing gap is not itself an obstruction: only c^2/g is priced.

The RH-96/RH-102 audit has 38 candidate events over thresholds $10^{-8}, 10^{-6}, 10^{-4}$. All local gap certificates are green; the primary five events all fit the stopped budget. The three aggressive chains reject three events and still preserve the endpoint gate, with worst stopped ratio 1.006033 versus unrestricted worst ratio 1.024921. The largest finite replay-debit/local-price ratio is 0.9741. This closes the uniform theorem's algebra and finite stress test, but not the physical all-level supply of gap and cross-energy exponents. No Stage A, Hilbert–Polya, zero identification, or Riemann Hypothesis result is claimed.

Keywords: gap-aware quotient; Ky Fan loss; stopped endpoint clock; spectral gap; adaptive Ritz enrichment; uniform price law.

MSC 2020: 47A75; 15A18; 65F15; 65G20; 37C30.

1 Introduction

RH-95 found that a fourth projected-cross direction can be numerically meaningless even when the corrected packet tail is stable. RH-96 replaced unstable direction recovery by a gap-weighted energy quotient, proving the local estimate

$$\Phi_r(H) - \Phi_r(A) \le \frac{\|C\|_{\mathrm{F}}^2}{\alpha - \beta}$$

under a retained-to-omitted gap [2]. The same paper also showed a crucial negative result: increasing the relative cutoff can leave every local certificate green while the recursive endpoint gate fails.

RH-102 solved the finite logical problem by assigning each proposed quotient an exact full-suffix debit and stopping before the accumulated absolute debit uses the endpoint slack [1]. What remained was the uniform question: what family-level quantity should be bounded so that a whole candidate supply can be admitted without replaying every possible future, and what remains safe when that supply is sparse?

The answer is a price law. A gap is not an independent resource; it appears only in the denominator of the quotient price c^2/g. This observation has three consequences:

1. a fixed positive gap is sufficient but stronger than necessary;
2. a collapsing gap can be harmless if the cross energy collapses faster;
3. a stopped clock is the correct fallback when the uniform price budget does not fit or the candidate supply terminates.

This paper proves those statements in one finite-dimensional theorem, derives the sigma-power criterion, and audits the full RH-96/RH-102 data.

2 Local gap price

For a positive Hermitian block matrix write

$$H = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix} \succeq 0, \qquad \Phi_r(X) = \sum_{j=1}^{r} \lambda_j^{\downarrow}(X). \tag{1}$$

The Ky Fan variational principle is

$$\Phi_r(X) = \max_{P=P^*=P^2,\, \operatorname{tr} P = r} \operatorname{tr}(PX).$$

Assume that the retained block has $\lambda_r^{\downarrow}(A) \ge \alpha$ and the omitted block satisfies $D \preceq \beta I$, with

$$g := \alpha - \beta > 0, \qquad c := \|C\|_{\mathrm{F}}\,. \tag{2}$$

Theorem 2.1 (Local gap-weighted quotient price)**.** *Under* (1)–(2)*,*

$$0 \le \Phi_r(H) - \Phi_r(A) \le \ell, \qquad \ell := \frac{c^2}{g}. \tag{3}$$

Proof. Let P be a rank-r projector on the full block space and write its blocks as P_{ij}. Put $t = \operatorname{tr} P_{22}$. Since $\operatorname{tr} P_{11} = r - t$ and the rth retained eigenvalue is at least α,

$$\operatorname{tr}(P_{11}A) \le \Phi_r(A) - \alpha t.$$

Also $\operatorname{tr}(P_{22}D) \le \beta t$. The projector identity gives $\|P_{12}\|_{\mathrm{F}}^2 \le t$. Hence

$$\operatorname{tr}(PH) - \Phi_r(A) \le -gt + 2c\sqrt{t} \le \frac{c^2}{g}.$$

Maximizing over P proves the upper bound. The lower bound follows by using projectors supported in the retained block. □

Remark 2.2 (The priced invariant). The theorem does not require a stable basis for the omitted space. It prices the whole weak block through the scalar ratio c^2/g. A lower bound on g alone is therefore not the correct uniform target.

3 Uniform all-candidate admission and stopped safety

Let R_- be a rigorous lower bound for a reference endpoint scale R, and let H_0^+ be an upper bound for the all-full baseline endpoint. For a gate $\Gamma > 1$, define the certified slack and allowance

$$S = (\Gamma R_- - H_0^+)_+, \qquad A = \eta S, \qquad 0 \le \eta < 1. \tag{4}$$

Suppose a candidate sequence contains N local quotients. At candidate i, let $\ell_i = c_i^2/g_i$ be the local price from Theorem 2.1, let $d_i \ge |\delta_i|$ be a certified propagated endpoint debit, and suppose

$$d_i \le K_i \ell_i. \tag{5}$$

Theorem 3.1 (Uniform gap-aware quotient law)**.** *Assume $S > 0$, every candidate has $g_i > 0$, and there are envelopes $K_i \le K$ and $\ell_i \le \bar{\ell}$. If*

$$NK\bar{\ell} \le A, \tag{6}$$

then the stopped clock accepts every candidate, and the terminal endpoint satisfies

$$|H_N - H_0| \le NK\bar{\ell} \le A, \qquad H_N < \Gamma R_- le \Gamma R. \tag{7}$$

If (6) *is not known, the same endpoint conclusion holds for the stopped prefix obtained by accepting only while the cumulative certified debits remain at most A and switching permanently to the full update after the first rejection.*

Proof. For the first assertion, the cumulative debit before any candidate is at most $NK\bar{\ell} \le A$, so the admission test accepts every candidate. The scalar hybrid contributions telescope and the triangle inequality gives $|H_N - H_0| \le \sum_i d_i \le NK\bar{\ell}$. Since $A = \eta(\Gamma R_- - H_0^+)$ with $\eta < 1$, the endpoint is strictly below ΓR_-. For the second assertion, the same telescoping argument holds for the accepted prefix, while the stopping rule enforces its cumulative debit bound. The later full suffix makes no further quotient debit. □

Remark 3.2 (What is new relative to a local certificate)*.* The local theorem controls ℓ_i at one update. The uniform law adds the candidate count N, the propagation factor K, and the endpoint slack. These are exactly the three quantities missing from a purely local gap argument. The stopped branch remains valid even when any one of them is too large for the no-stop inequality.

4 Sigma-power criterion

The uniform law becomes a concrete all-level target after recording signed powers. Let $L(\sigma)$ denote a generic product of polylogarithmic factors.

Theorem 4.1 (Power-priced quotient criterion)**.** *Suppose, for all sufficiently small σ, that*

$$N_\sigma \le C_N L_N(\sigma), \tag{8}$$
$$K_\sigma \le C_K \sigma^{-\kappa} L_K(\sigma), \tag{9}$$
$$c_{\sigma,i} \le C_c \sigma^{\chi} L_c(\sigma), \tag{10}$$
$$g_{\sigma,i} \ge C_g \sigma^{\gamma} / L_g(\sigma), \tag{11}$$
$$S_\sigma \ge C_S \sigma^{s} / L_S(\sigma). \tag{12}$$

Then the total uniform price obeys

$$N_\sigma K_\sigma \sup_i \frac{c_{\sigma,i}^2}{g_{\sigma,i}} \le C\, \sigma^{2\chi-\gamma-\kappa} L_N L_K L_c^2 L_g, \tag{13}$$

for a constant C depending only on the displayed constants. If

$$2\chi - \gamma - \kappa > s, \tag{14}$$

then the no-stop budget in (6) *holds for all sufficiently small σ. At equality, it holds provided the constant and polylogarithmic factors fit the safety fraction.*

Proof. Square the coupling bound, divide by the gap lower bound, and multiply by the candidate-count and propagation envelopes. This gives (13). Comparing it with the slack lower bound in (12) proves the strict-power assertion; equality leaves only the constants and logarithms. ☐

The theorem is deliberately phrased with a gap decay exponent γ rather than a uniform fixed gap. It permits $g_\sigma \to 0$ whenever c_σ vanishes faster.

Proposition 4.2 (Gap collapse is not the invariant)**.** *Let $g_\sigma = \sigma^\gamma$ and $c_\sigma = \sigma^\chi$ with $\gamma > 0$. Then*

$$\frac{c_\sigma^2}{g_\sigma} = \sigma^{2\chi-\gamma}. \tag{15}$$

In particular, for $\gamma = 1$, the family $c_\sigma = \sigma^{5/4}$ has price $\sigma^{3/2}$ even though the gap collapses, while $c_\sigma = \sigma^{1/2}$ has constant price one.

Proof. Direct substitution gives (15). The two stated cases have exponents $2(5/4) - 1 = 3/2$ and $2(1/2) - 1 = 0$. ☐

5 Five-anchor stopped-clock audit

The audit independently reads RH-96 local gap events and RH-102 stopped hybrid chains. The local price is recomputed as c^2/g from the stored coupling and gap bounds. For each accepted or rejected hybrid event, the stored propagated debit is divided by the recomputed local price to obtain a finite replay multiplier. This checks the three inputs to Theorem 3.1: local price, propagation price, and endpoint budget.

Table 1: Local prices and stopped endpoint outcomes over the three thresholds.

threshold	candidates	accepted/rejected	min gap	max c^2/g	stopped worst
10^{-8}	5	5/0	$2.222\,10^{-12}$	$4.279\,10^{-13}$	1.001173
10^{-6}	11	10/1	$2.189\,10^{-12}$	$1.395\,10^{-9}$	1.001173
10^{-4}	22	20/2	$2.213\,10^{-12}$	$5.029\,10^{-8}$	1.006033

There are 38 candidate events across the three independent threshold chains; 35 are accepted and three are rejected. Every one of the 38 local gap certificates is green. The primary threshold has no rejection and uses only 0.102% of its allowance. The largest finite replay multiplier is 0.9741, so the observed endpoint prices are no larger than their local gap-price envelope on these frozen chains.

The aggressive thresholds show why the stopped branch matters. Their unrestricted worst endpoint/reference ratios are 1.024921 and 1.014092, whereas the stopped ratios are 1.001173 and 1.006033. Local validity alone does not imply global composition, but the stopped law restores the endpoint gate.

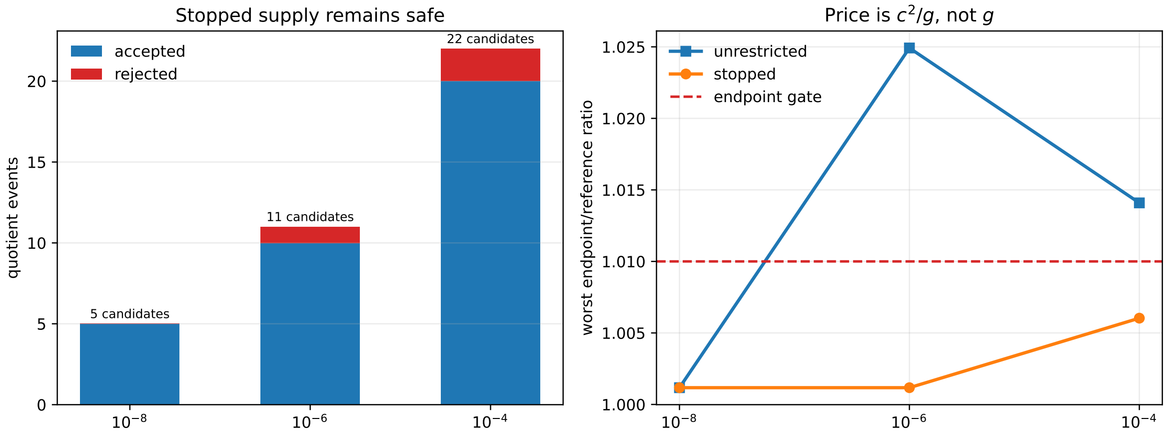

Figure 1: Candidate supply and endpoint outcomes for the three thresholds. The primary price supply fits without stopping; aggressive supplies are truncated before the endpoint budget is exhausted.

For a separate ratio sanity check, the audit also evaluates the two families in Proposition 4.2 at $\sigma = 10^{-1}, \ldots, 10^{-6}$. The good family has price power $3/2$ despite a vanishing gap; the bad family has constant price. This is the finite numerical signature of the power criterion, not an estimate of the physical production exponents.

6 Route consequence and claim boundary

RH-106 changes the uniform quotient question from "is there a gap?" to three measurable conditions:

1. a local price envelope $\sup c^2/g$;
2. a propagation envelope K and candidate count N;
3. an endpoint slack lower bound.

The power condition $2\chi - \gamma - \kappa \geq s$ is a sufficient no-stop route, while the stopped theorem remains valid if the inequality fails. This is a genuine narrowing of the all-level gate, but it is not its proof: the physical folded-Gaussian family has not yet supplied uniform exponents $(\chi, \gamma, \kappa, s)$ or a replay-free K.

The current route ledger is therefore:

- RH-104 identifies the exact physical finite-prefix target;
- RH-105 closes signed observation–residual algebra;
- RH-106 closes the abstract uniform price/stopped-safety law;
- physical prefix, residual, upstream, and quotient exponents remain open all-level inputs.

No statement here constructs a Hilbert–Polya operator, proves a $T \log T$ law or prime-power trace formula, identifies zeta zeros, or proves the Riemann Hypothesis.

7 Reproducibility

The directory contains the quotient-price helpers, full and smoke audits, figures, tests, and a hash-checked archive. Run:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \\
  experiments/build_quotient_audit.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \\
  experiments/build_quotient_audit.py --smoke
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \\
  experiments/make_figures.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
```

The archive scripts record the exact RH-96/RH-102 inputs and verify all publication hashes.

References

[1] Liang Wang. Stopped hybrid quotient clocks. Preprint and software repository, RH-102, 2026.

[2] Liang Wang. Gap-weighted weak-mode quotients. Preprint and software repository, RH-96, 2026.